Atypical functional hierarchy of the human cerebellum in autism

Author list

Wenjing Zhu[1], Yu Zhang[1], Ruoqi Yang[1], Shuo Lv[1], Tan Gao[1], Xinyu Wu[1], Tiantian Liu[2,3], Chuyang Ye[3,4], Guoyuan Yang[1,2,3,*], Tianyi Yan[2,3,*]

Affiliations

[1] School of Interdisciplinary Science, Beijing Institute of Technology, Beijing,100081, China

[2] School of Medical Science and Engineering, Beijing Institute of Technology, Beijing,100081, China

[3] Beijing Key Laboratory of Brain-inspired Neural Engineering, Beijing,100081, China

[4] School of Integrated Circuits and Electronics, Beijing Institute of Technology, Beijing,100081, China

*Corresponding authors: Guoyuan Yang, email: yanggy@bit.edu.cn

Tianyi Yan, email: yantianyi@bit.edu.cn

Abstract

Cerebellar abnormalities have been increasingly implicated in autism spectrum disorder (ASD), yet whether these alterations reflect disruptions in large-scale cerebellar functional hierarchy remains unclear. Here, we applied manifold learning to characterize cerebello–cortical functional connectivity gradients using multisite resting-state functional magnetic resonance imaging data from the Autism Brain Imaging Data Exchange (ABIDE). Compared with healthy controls, individuals with ASD exhibited atypical organization of the principal cerebellar gradient spanning sensory and transmodal association regions, particularly those associated with the default mode network. Individualized gradient patterns distinguished ASD from healthy controls with high accuracy (97.47%, permutation $p < 0.001$), with the ventral attention and default mode networks contributing prominently to classification. Exploratory analyses revealed network-specific associations between cerebellar gradient alterations and clinical symptom severity, with ventral attention and sensorimotor networks linked to autistic traits and restricted and repetitive behaviours, whereas dorsal attention and sensorimotor networks were primarily associated with social impairment. Furthermore, spatial transcriptomic analyses revealed that ASD-related cerebellar gradient alterations were associated with gene expression patterns enriched for synaptic signalling, neurodevelopment and cellular regulatory processes. The principal findings were further replicated in an independent validation cohort. Together, these findings reveal atypical cerebellar functional hierarchy in ASD and highlight its potential relevance to clinical heterogeneity and underlying molecular organization.

Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental condition characterized by persistent difficulties in social communication and interaction, together with restricted and repetitive behaviours(1). Building on foundational theories of cortical information processing, ASD is increasingly conceptualized as a disorder of large-scale brain organization, characterized by atypical integration along the sensory-to-association cortical hierarchy and a disruption of its functional hierarchical architecture(2-10). These cortical hierarchy abnormalities are further supported by converging microcircuit and cellular–molecular evidence(11, 12). Although most neuroimaging work has focused on the cortex, converging structural, functional, genetic, developmental and circuit-level evidence suggests that cerebellar dysfunction may represent another major contributor to ASD pathophysiology(13-20). Moreover, ASD-risk genes are expressed in cerebellar circuits, and early cerebellar injury has been identified as one of the strongest non-heritable risk factors for later ASD-related outcomes(14, 21). Furthermore, experimental and clinical evidence implicates cerebellar regions, including right Crus I and the posterior vermis, in the regulation of social behaviour and restricted and repetitive behaviours through cerebello–prefrontal circuits(22, 23). Despite this evidence, whether ASD is associated with atypical cerebellar functional hierarchy at the neural systems level remains unknown.

Recent advances in systems neuroscience have enabled gradient-based approaches using functional magnetic resonance imaging (fMRI) to characterize functional hierarchy as a low-dimensional principle of brain organization(24, 25). In the cortex, the principal functional gradient captures a sensory-to-association axis extending from primary unimodal regions to transmodal association networks, providing a compact representation of macroscale functional hierarchy(24, 25). This architecture is thought to support the integration of abstract concepts, cognition and behaviour(26, 27). Lifespan studies further suggest that cortical gradient architecture follows continuous and nonlinear trajectories from infancy to late adulthood, recapitulating known developmental timing within a comprehensive normative chart of functional organization(28, 29). At the microscopic scale, this hierarchy is supported by interacting and partially opposing molecular gradients that mirror cortical organization, consistent with competing developmental programs shaping the sensorimotor–association axis and undergoing postnatal refinement(30, 31). In ASD, cortical gradient studies have revealed a system-level transition in functional hierarchy, characterized by compression of the principal sensory-to-transmodal gradient and reduced differentiation between sensory and higher-order networks(2, 32). By capturing macroscale organizational features across the entire cortical mantle, cortical gradients provide a principled framework for examining associations between hierarchical network architecture and a broad spectrum of autism-related behavioral phenotypes, including social cognition, communication, and restricted or repetitive interests(2). However, although the cerebellum likewise exhibits a prominent gradient architecture progressing from sensory-to-transmodal territories, atypical cerebellar functional gradients in autism remain largely unexplored(33).

Growing interest in modelling individual differences in brain function has motivated analytic frameworks that link intrinsic neural organization to stable biological and behavioural traits(34-36). Functional connectivity profiles can identify individuals and predict cognitive ability, supporting the view that intrinsic brain organization carries reliable person-specific information(34). Individual-specific cortical network topography

and functional gradients have been shown to predict variability in cognition, personality and emotion(37, 38). Emerging cerebellar studies further suggest that lobule-specific structure, functional gradients and cerebello–cortical connectivity are sensitive to behavioural variation across motor, cognitive and social domains(33, 39-41). These findings raise the possibility that individualized atypical cerebellar hierarchy may provide a clinically meaningful phenotype for autism, but it remains unclear whether cerebellar gradient abnormalities are associated with symptom severity and whether distinct cerebellar networks contribute differentially to social impairment, autistic traits, interoceptive and interpersonal phenotypes.

Here, we used manifold learning to characterize cerebello–cortical interactions derived from functional connectomes in the multisite Autism Brain Imaging Data Exchange dataset (ABIDE I and II). We mapped cerebellar functional gradients at both group and individual levels, quantified autism-related alterations in unimodal-to-transmodal gradient transitions, and tested whether gradient features could distinguish individuals with autism from healthy controls. We then related individualized gradient abnormalities to symptom severity and examined their association with core clinical autistic phenotypes. Furthermore, to explore the potential genetic underpinnings of these atypical functional patterns, we used a multivariate approach to relate functional hierarchy abnormalities to cerebellar gene-expression patterns. Finally, to strengthen the generalizability of our findings, we examined the reproducibility of these findings in the ABIDE II multisite validation cohort. This framework provides a multiscale account of cerebellar functional hierarchy in autism and highlights the cerebellum as a richly structured component of whole-brain functional organization.

Results

Two independent cohorts from the Autism Brain Imaging Data Exchange (ABIDE I and ABIDE II; http://fcon_1000.projects.nitrc.org/indi/abide)(42, 43) were analyzed, including a discovery cohort from ABIDE I and an independent replication cohort from ABIDE II (see Methods). In both datasets, only imaging sites that recruited both children and adults with ASD and healthy controls (HCs) were included(2). After rigorous imaging quality control, including exclusion of participants with poor-quality structural MRI, failed surface reconstruction, or excessive head motion during resting-state fMRI (mean framewise displacement > 0.3 mm), the final discovery cohort of ABIDE I comprised 243 participants (119 ASD and 124 HCs) from three sites (NYU, USM, and PITT), whereas the replication cohort comprised 109 participants (55 ASD and 54 HCs) from three independent sites (TCD, NYU, and IP). ASD and HC participants were well matched for age and head motion both across the entire cohort and within each imaging site (all p > .05). Detailed demographic characteristics are summarized in **Table 1**.

Altered cerebellar functional gradients in ASD

We first characterized the principal organizational axis of cerebello-cerebral functional connectivity using Laplacian Eigenmap–based gradient analysis(24, 44). Specifically, we (i) computed a correlation matrix between 17,853 cerebellar voxels and 1,483 cortical ROIs using Pearson's correlation, (ii) constructed a 17,853 × 17,853 affinity matrix using cosine similarity between the cerebellar connectivity profiles, and (iii) applied the Laplacian Eigenmap algorithm to derive low-dimensional gradient components.

Consistent with previous study(45), the principal gradient revealed a continuous transition from primary sensorimotor territories to transmodal regions in both ASD and HCs (**Fig. 1A**). The principal gradient (Gradient 1) explained the largest proportion of variance in cerebellar functional connectivity in both groups (ASD: 31.8%; HCs: 30.7%), whereas the second gradient (Gradient 2) accounted for substantially less variance (ASD: 16.2%; HCs: 17.0%; **Supplementary Fig. S1**).

Despite this preserved macroscale organizational axis, individuals with ASD exhibited a contraction at the transmodal end of the principal gradient distribution (**Fig. 1B**), indicating a compressed functional hierarchy. At the network level, we compared the ASD and HC group-averaged gradient maps using two-sample *t*-tests across spatially corresponding voxels within each functional network, with false discovery rate (FDR) correction applied across networks. Compared with the HC group, the ASD group showed significantly higher principal gradient values in the ventral attention (VAN), frontoparietal (FPN), and dorsal attention (DAN) networks, but lower gradient values in the visual (VN), somatomotor (SMN), and default mode (DMN) networks (FDR-corrected $q < 0.05$; **Fig. 1C and Supplementary Table S1**). The VAN showed the largest standardized between-group difference (Cohen's $d = 0.77$), whereas the remaining networks exhibited moderate-to-small effect sizes.

Examination of the first two embedding dimensions further revealed shifts in the spatial positioning of functional network centroids within gradient space (**Fig. 1D**). Compared with HCs, the SMN centroid was displaced primarily toward higher Gradient 2 values, whereas the VAN centroid shifted toward both higher Gradient 1 and higher Gradient 2 values. These shifts suggest altered relative positioning of these functional networks within the cerebellar functional gradient space.

To further characterize these findings across age groups, we performed post-hoc analyses separately in children (<18 years) and adults (≥18 years). Across the six functional networks, the patterns and magnitudes of ASD–HC differences were broadly similar between children and adults, with no significant Group × Age interactions in any network (all $p > 0.05$; **Supplementary Fig. S2**). Although the SMN showed a larger effect in children (Cohen's $d = -0.68$) than in adults (Cohen's $d = -0.18$), the Group × Age interaction did not reach statistical significance ($p = 0.064$). Together, these findings suggest that ASD-related alterations in cerebellar principal gradient are broadly consistent between children and adults, with no significant evidence of age-group differences in the magnitude of these effects.

Importantly, the overall principal gradient architecture was highly consistent in the independent ABIDE II cohort (**Supplementary Fig. S3**). Similar to the discovery cohort, ASD showed reduced gradient values within the DMN together with elevated values in attention-related association networks, including the VAN, FPN, and DAN.

Together, the two independent cohorts converged on a common finding that ASD is characterized by compression of the cerebellar functional hierarchy, despite modest cohort-specific variation in network alterations.

Within cerebellar connectivity shows convergent gradient alterations in ASD

To further determine whether the atypical gradient organization observed in cerebello-cortical connectivity was similarly manifested within the cerebellum itself, we next examined intrinsic cerebellar functional gradient patterns. To this end, we derived functional gradients from cerebellum-to-cerebellum connectivity, providing a measure of intrinsic cerebellar organization without direct contributions from cortical connectivity.

The intrinsic cerebellar gradients revealed broad features of cerebellar functional organization, including a sensorimotor-to-transmodal axis in principal gradient (**Supplementary Fig. S4A**). Individuals with ASD showed a contraction of the principal gradient distribution relative to HCs, accompanied by significantly lower gradient values in the VN, SMN, and DMN and higher values in the DAN, VAN, and FPN (FDR-corrected $q < 0.05$; **Supplementary Fig. S4B–C**). Network centroids also showed systematic shifts within gradient space in ASD (**Supplementary Fig. S4D**).

Together, the partial convergence between intrinsic cerebellar and cerebello-cortical gradients further supports the robustness of cerebellar functional reorganization in ASD, suggesting that these alterations reflect intrinsic functional abnormalities of the cerebellum rather than being solely driven by atypical cortical connectivity.

Statistical comparison of gradient and their cognitive associations in ASD

Voxel-wise comparisons of individual principal gradient values identified widespread regional alterations across the cerebellum (FDR-corrected $q < 0.05$; **Fig. 2A**). Compared with HCs, individuals with ASD exhibited significantly lower gradient values predominantly within regions assigned to the DMN (37.1%) and VN (25.3%), whereas significantly higher gradient values were primarily observed within the FPN (46.8%).

Meta-analytic decoding demonstrated that cerebellar regions exhibiting altered principal gradient values in ASD were predominantly associated with transmodal association and sensory-related functions, including coordination, timing, working memory, and movement (**Fig. 2B**). Notably, these cognitive profiles were highly similar for regions showing increased and decreased gradient values, suggesting that ASD-related gradient alterations are embedded within a common sensorimotor and association functional hierarchy rather than representing distinct cognitive systems. For details, see **Supplementary Table S2.**

Individual variability of cerebello-cortical gradients in ASD

Beyond regional mean differences, we next examined whether ASD was associated with altered individual variability in cerebello-cortical gradient organization. Individuals with ASD showed significantly lower similarity to the group-average principal gradient than HCs (**Fig. 2C**), indicating reduced consistency of cerebellar functional organization across individuals ($p < 0.01$).

Global gradient dispersion showed a trend toward increased variability in ASD ($p = 0.057$; **Fig. 2C**). We further examined dispersion within each canonical functional network. This analysis revealed significantly reduced dispersion in the VN and significantly increased dispersion in the VAN and FPN in ASD relative to HCs (all $p < 0.05$; **Fig. 2D**). These findings indicate that alterations in the spatial dispersion of cerebello-cortical functional

organization in ASD are network-specific, with opposite patterns of dispersion in sensory and association-related systems.

Together, these findings suggest that ASD-related cerebellar abnormalities involve not only shifts in regional gradient values but also increased inter-individual variability within higher-order association and sensory networks.

Functional gradient–based machine learning classification between ASD and HCs

Following the identification of ASD-related alterations in functional gradients, we next evaluated whether individualized functional gradients could discriminate individuals with ASD from HCs. Individual principal gradient maps derived from the ABIDE I dataset were used to train a linear support vector machine (SVM) classifier. In each of 100 repetitions, participants were randomly divided into training (80%) and held-out testing (20%) sets while preserving the ASD-to-HC ratio. The classifier achieved a mean accuracy of 97.47% on the held-out testing sets (permutation test, $p < 0.001$; **Fig. 3A**), with a mean area under the receiver operating characteristic curve (AUC) of 0.995.

To characterize the spatial contribution of cerebellar functional gradients to classification performance, we computed Haufe-transformed weight maps from the trained SVM models (**Fig. 3B**). Positive weights indicate cerebellar regions whose gradient values were positively associated with the ASD group (i.e., higher values in ASD than in HCs), whereas negative weights indicate regions positively associated with HCs (i.e., higher values in HCs than in ASD). To further evaluate the network-level contribution of functional gradient, Haufe-transformed weights were summarized within each of the six canonical cerebellar functional networks (**Fig. 3C**). Transmodal association networks, particularly the VAN and DMN, showed the largest positive and negative contributions to ASD classification, respectively, whereas the FPN also contributed prominently to the model. Then, we examined the spatial correspondence between the Haufe-transformed weight map and the group-level principal gradient difference *t*-map. We found that the two maps showed a strong spatial correlation (Spearman's $r = 0.96$, Moran permutation $p < 0.001$; **Fig. 3D**), indicating that the classifier primarily corresponded to case–control gradient alterations.

To further assess the cross-site generalizability of the gradient-based classification approach, we performed leave-one-site-out validation within the ABIDE I cohort. In each analysis, participants from one site were held out as an independent test set, while participants from the remaining sites were used for model training. The classifier showed consistently high performance across the three held-out sites, with mean accuracy of 93.78% for USM, 97.03% for NYU, and 97.45% for PITT (**Supplementary Fig. S5**). These results further supported the cross-site generalizability of the gradient-based classifier.

We further examined the reproducibility of the gradient-based classification approach in the independent ABIDE II cohort. Using the principal gradient derived from ABIDE II, we independently trained and evaluated the classifier within this cohort. The model again achieved high classification performance (mean accuracy = 89.29%, $p < 0.001$; AUC = 0.960; **Supplementary Fig. S6**). Moreover, the spatial correspondence between the Haufe-transformed weight map and the group-level principal gradient difference map was preserved ($r = 0.95$, Moran permutation $p < 0.001$).

Together, these findings demonstrate that individualized cerebellar functional gradients can reliably distinguish individuals with ASD from HCs, with robust classification performance across sites and independent cohorts. Moreover, the strong spatial correspondence between classification weights and group-level gradient alterations suggests that cerebellar functional gradients capture meaningful ASD-related differences in functional organization.

Associations between network-specific gradients and symptom severity

To further explore potential associations between network-specific cerebellar principal gradient values and symptom heterogeneity in ASD, we conducted three exploratory canonical correlation analyses (CCA) relating six network-specific principal gradient values to three predefined behavioral domains (core autistic traits, restricted and repetitive behaviors (RRBs), and social impairment) in the ABIDE II cohort (see Methods). All three CCA models remained significant after FDR correction (all $q < 0.05$; **Fig. 4**), indicating significant multivariate associations between cerebellar functional gradients and distinct autistic behavioral phenotypes.

The first CCA model revealed a significant association between principal gradient and core autistic traits ($r = 0.83$, permutation $p = 0.006$, FDR-corrected $q = 0.009$; **Fig. 4A**). The corresponding canonical mode was primarily driven by the VAN and SMN, while the DMN made the largest contribution in the negative-weight direction. The second model identified a significant association with RRBs ($r = 0.75$, permutation $p = 0.028$, FDR-corrected $q = 0.028$; **Fig. 4B**), with the largest contributions also arising from the VAN and SMN, while the FPN made a relatively greater contribution in negative-weight. The third model demonstrated a significant association between principal gradient and social impairment ($r = 0.77$, permutation $p = 0.004$, FDR-corrected $q = 0.009$; **Fig. 4C**), in which the canonical mode was primarily driven by the DAN and SMN with the DMN making the primary contribution to the negative weights.

Notably, the VAN contributed most strongly to the canonical modes associated with core autistic traits and RRBs, whereas the DAN was the dominant contributor to the canonical mode associated with social impairment. Across all three models, positive canonical correlations indicated that higher network-specific principal gradient loadings were associated with greater behavioral impairment.

Transcriptomic signatures of cerebellar gradients in ASD

To investigate the molecular architecture underlying ASD-related alterations in cerebellar principal gradients, we performed partial least squares (PLS) analysis between regional gene expression profiles from the Allen Human Brain Atlas(46, 47) and the regional case–control gradient difference *t*-map (**Fig. 5A and B**). The first latent component (PLS1) explained 60.17% of the variance in the regional t-statistic map, and its significance was supported by permutation testing ($p_{perm} = 0.003$). The spatial distribution of PLS1 scores was significantly correlated with regional case–control differences in principal gradients ($r = 0.78$, $p_{moran} < 0.001$; **Fig. 5C**), indicating a strong spatial correspondence between the transcriptional pattern and ASD-related cerebellar gradient alterations.

Bootstrap analysis identified genes with stable positive and negative PLS1 loading weights (**Fig. 5D**). Genes were ranked according to their normalized PLS1 weights, and

genes with $Z > 2.58$ or $Z < -2.58$ were selected as $PLS1^+$ and $PLS1^-$ gene sets, respectively. Functional enrichment analysis of the $PLS1^+$ genes revealed prominent involvement of synaptic signaling and neurodevelopmental processes, including modulation of chemical synaptic transmission, synaptic signaling, and neuron development. These genes were also enriched in processes related to chromatin remodeling, mRNA metabolism, intracellular signaling, cellular stress responses, and membrane organization (**Fig. 5E and F**). In contrast, the $PLS1^-$ genes were primarily enriched in processes related to cell adhesion, cell migration, cytoskeletal organization, and intracellular and vesicle-mediated transport, while also showing enrichment for neuronal and central nervous system development (**Supplementary Fig. S7**). Together, these findings suggest that the transcriptomic pattern associated with ASD-related cerebellar gradient alterations involves molecular programs related to synaptic function and neurodevelopment, together with broader cellular regulatory, structural, and transport processes.

Discussion

Using two independent multisite cohorts, we identified an atypical organization of cerebello-cortical functional hierarchy in ASD. Although the principal cerebellar gradient retained its canonical sensorimotor-to-transmodal organization, ASD was characterized by a contraction of its transmodal end, with the most consistent alterations involving cerebellar territories linked to the DMN network. These abnormalities were accompanied by reduced similarity to the group-level gradient architecture and greater network-specific dispersion, suggesting that ASD involves both a systematic shift and increased interindividual heterogeneity in cerebellar functional organization. Individual gradient profiles further showed diagnostic-group discrimination, symptom-related associations, and spatial correspondence with cerebellar transcriptional patterns. Together, these findings advance our understanding of the cerebellar contributions to ASD pathophysiology and highlight cerebellar functional gradients as potential neuroimaging markers of clinical heterogeneity.

The principal cerebellar gradient recapitulates the progressive organization of cerebello-cortical connectivity from motor and sensory territories toward regions associated with executive, social-cognitive, and internally directed processing(33, 48, 49). In ASD, reduced gradient values within cerebellar DMN territories, together with elevated values in the FPN and attention-related networks, indicate an alteration in cerebellar functional hierarchy. The contraction of the transmodal end of the principal gradient suggests diminished differentiation along the sensorimotor-to-association axis. This finding parallels cortical gradient compression previously reported in ASD(2), supporting the possibility that reduced sensory–association differentiation is a distributed property of cerebello-cerebral organization(50, 51). Importantly, this pattern was consistently observed in intra-cerebellar functional connectivity, indicating that hierarchical disruption in ASD is intrinsic to the cerebellum rather than merely a redundant reflection of cortical abnormalities. Given the reciprocal connections of posterior cerebellar regions with prefrontal and parietal association cortices and their functional coupling with the DMN, altered hierarchical organization in these regions may disrupt the integration of sensory information with abstract, contextual, and internally generated representations, potentially contributing to impairments in executive control, cognitive flexibility, and social cognition in ASD(33, 49, 52, 53). Consistent with this interpretation, our meta-analytic decoding

revealed that cerebellar regions exhibiting altered gradient values were associated with both sensorimotor and higher-order cognitive functions, including coordination, timing, and working memory. Notably, regions showing increased and decreased gradient values exhibited highly similar functional profiles, suggesting that these alterations occur within a shared sensorimotor–association hierarchy rather than distinct cognitive systems. Together, these findings suggest a potential link between disrupted cerebellar functional hierarchy and the co-occurrence of sensory and cognitive impairments in ASD.

The case–control statistical map additionally showed opposite-signed effects across portions of the left and right cerebellar hemispheres. This spatial pattern raises the possibility that ASD-related gradient abnormalities involve altered cerebellar lateralization in addition to global hierarchical compression. Functional asymmetry is an intrinsic feature of cerebello-cortical organization, partly reflecting crossed projections between the cerebellum and lateralized cerebral systems(2, 54-56). Right cerebellar regions, for example, are preferentially coupled with left-lateralized language and verbal-association cortices, whereas left cerebellar territories show complementary relationships with right-hemisphere attentional and visuospatial systems. Opposing case–control effects in approximately corresponding hemispheric regions may therefore reflect an altered balance between lateralized cerebro-cerebellar loops. This interpretation is relevant to ASD, in which atypical lateralization has been reported for language, social cognition, and attentional processing(57, 58). Nevertheless, the observed pattern should not be interpreted as definitive evidence of abnormal asymmetry without a formal diagnosis-by-hemisphere interaction. Because gradient signs are mathematically arbitrary before alignment, lateralization analyses also require a common reference orientation and replication across cohorts. The present findings therefore motivate targeted analyses using mirrored cerebellar regions and individual asymmetry indices.

Beyond group-level differences, individuals with ASD showed reduced similarity to the group-average cerebellar gradient, accompanied by network-specific alterations in dispersion: reduced dispersion in the VN and increased dispersion in the VAN and FPN. These findings suggest that cerebellar hierarchical abnormalities in ASD involve not only systematic shifts in functional organization but also alterations in the spatial variability of individual gradient patterns, with distinct effects across sensory and association-related systems. Previous studies have demonstrated substantial individual variability in cerebellar network topography(53), while idiosyncratic cortical connectivity patterns and heterogeneous neuroanatomical deviations have been widely reported in ASD(59-61). Our findings extend these observations to cerebellar functional hierarchy, highlighting network-specific alterations in individual variability rather than a uniform change across the cerebellum in ASD. These network-specific alterations may reflect individual variability in cerebellar functional hierarchy, with heterogeneous regional deviations potentially converging on common functional systems in ASD(62). Future studies using normative modeling may help characterize individualized cerebellar deviations and their clinical relevance(63).

The associations between cerebellar functional gradients and symptom severity further suggest that disrupted cerebellar hierarchy may contribute to the behavioral heterogeneity of ASD. Specifically, gradient alterations in the VAN and SMN contributed most strongly to multivariate associations with core autistic traits and restricted and repetitive behaviors. The ventral attention system supports the detection of behaviorally relevant stimuli and the reorienting of attention in response to changing environmental demands(64). Given the

cerebellum's established role in predictive processing and adaptive behavioral control(65), altered hierarchical organization across these networks may compromise the flexible updating of sensory predictions and behavioral responses, potentially contributing to the behavioral rigidity and repetitive patterns characteristic of ASD(66). In contrast, social impairment was predominantly associated with gradient features in the DAN and SMN, with an opposing contribution from the DMN. The dorsal attention system supports goal-directed selection of external information, whereas the DMN is implicated in internally directed processing and mentalizing. Successful social interaction requires the continuous integration of socially relevant sensory cues with contextual information and predictions about others' intentions(67, 68). Disrupted hierarchical coordination across these systems may therefore impair the ability to translate dynamic social cues into contextually appropriate responses. This finding is consistent with previous evidence linking cortical hierarchical disruption to both sensory and socio-cognitive symptoms(2), as well as experimental studies implicating cerebellar circuits in social and repetitive behaviors(68, 69). Together, these findings suggest that network-specific alterations in cerebellar functional hierarchy may provide a common neurobiological framework linking atypical sensory processing, behavioral inflexibility, and social difficulties in ASD. It should be noted, however, that these findings are based on a relatively small sample and should therefore be considered exploratory, warranting further validation of the gradient–symptom associations in larger cohorts.

The imaging–transcriptomic analysis provided a potential molecular context for ASD-related alterations in cerebellar functional hierarchy. Genes with positive PLS1 loadings were enriched for synaptic signaling, chemical synaptic transmission, and neuronal development, whereas genes with negative loadings were primarily associated with cell adhesion, cytoskeletal organization, and intracellular transport. These findings are consistent with converging genetic evidence implicating synaptic dysfunction and disrupted neurodevelopmental processes in ASD(70, 71). Experimental studies further demonstrate that perturbations of cerebellar circuits can induce social deficits and repetitive behaviors, highlighting the relevance of cerebellar dysfunction to ASD pathophysiology(69, 72). Nevertheless, these associations reflect spatial correspondence with normative gene-expression patterns rather than disease-related transcriptional changes and should therefore be interpreted as hypothesis-generating(73).

This work has several limitations that point to promising directions for future research. First, multisite differences in acquisition and clinical characterization may persist despite harmonization and replication. Second, the broad age range and cross-sectional design preclude conclusions regarding the developmental origin of gradient compression. Nonlinear age-by-diagnosis analyses will be required to determine whether these alterations represent delayed maturation, atypical developmental trajectories, or stable differences. Third, sex composition differed between cohorts, and potential effects of medication and comorbidity could not be fully evaluated. Fourth, the behavioral analyses were based on relatively small samples with partially overlapping measures.

Collectively, our findings reveal a disruption of cerebellar functional hierarchy in ASD, characterized by reduced sensory–transmodal differentiation and altered organization of default-mode, frontoparietal, and attention-related networks. These alterations, together with individual-level deviations and their associations with clinical symptoms and transcriptional patterns, highlight the relevance of cerebellar hierarchical organization to ASD pathophysiology. Our study extends current models of atypical functional hierarchy

beyond the cerebral cortex and provides new insights into the neurobiological basis of clinical heterogeneity in ASD.

Methods

Participants

Two independent sample sets, namely a discovery cohort and a replication cohort, were selected separately from two publicly available releases of the Autism Brain Imaging Data Exchange (ABIDE I and II)(42, 43).

The discovery sample was derived from an existing ABIDE I subsample reported in prior work(42, 43). In brief, we only retained research sites recruiting both children and adults with autism spectrum disorder (ASD) and healthy controls (HCs), with a minimum of 10 participants per diagnostic group. This initial screening yielded a total of 258 subjects, including 128 individuals with ASD and 130 HCs. Considering the scarcity of female participants across ABIDE I, only male subjects were included in subsequent analyses(2). Participants with an IQ score >70 were retained for subsequent analyses. We performed rigorous imaging quality control, removing participants presenting poor-quality T1-weighted structural scans, failed cortical surface reconstruction, or excessive head movement during resting-state functional MRI (rs-fMRI; more than 0.3 mm framewise displacement (FD) were excluded). After quality filtering, the final discovery sample contained 243 participants recruited from three sites: (1) NYU Langone Medical Center (NYU; 58 ASD, 66 HCs); (2) University of Utah, School of Medicine (USM; 43 ASD, 38 HCs); (3) University of Pittsburgh, School of Medicine (PITT; 18 ASD, 20 HCs).

The replication cohort was sampled from ABIDE II(43). Overall inclusion criteria were consistent with those applied to the discovery cohort, except that female participants were also eligible given the relatively higher proportion of females within ABIDE II. In addition, the IQ-based exclusion criterion applied to ABIDE-I was not imposed on ABIDE-II because of the substantial amount of missing IQ data and the relatively small sample size, in order to retain as many eligible participants as possible. All other quality control procedures were kept consistent with those used for the discovery cohort. A preliminary pool of 177 participants was identified across three participating sites. Following identical quality control procedures adopted for the discovery dataset, 109 individuals (18 females) remained for replication analyses: (1) Trinity Centre for Health Sciences, Trinity College Dublin (TCD; 13 ASD, 17 HCs); (2) NYU Langone Medical Center (NYU; 28 ASD, 22 HCs); (3) Institut Pasteur/Robert Debré Hospital (IP; 14 ASD, 15 HCs).

All ASD diagnoses were confirmed via structured or unstructured face-to-face clinical interviews. Diagnoses encompassing autistic disorder, Asperger's syndrome, and pervasive developmental disorder not otherwise specified (PDD-NOS) were determined through expert clinical judgment aided by gold-standard diagnostic assessments, including the Autism Diagnostic Observation Schedule (ADOS)(74) and/or the Autism Diagnostic Interview-Revised (ADI-R)(75). These assessments quantify impairments across three core symptom domains: reciprocal social interaction, verbal and communicative ability, and restricted, repetitive patterns of behaviors and interests. Full-scale, verbal, and performance IQ measurements were acquired using the WASI, WAIS-III, and/or WISC-

III according to site-specific assessment protocols. Control participants had no lifetime history of psychiatric illness. Within each individual site, HCs were statistically matched to ASD participants on chronological age and rs-fMRI head motion. No significant group differences in age or head motion were detected between ASD and HC groups in either ABIDE I or II. Original data collection protocols for ABIDE I and II received ethical approval from local institutional review boards (IRBs), and all shared data were fully de-identified: all HIPAA-protected health identifiers were eliminated, and facial features on structural MRI scans were removed to prevent subject re-identification.

Data acquisition and preprocessing

High-resolution T1-weighted (T1w) structural MRI and rs-fMRI data were acquired on 3T MRI scanners (Siemens or Philips) across all participating sites. Detailed acquisition parameters for each site are available through the ABIDE repository (http://fcon_1000.projects.nitrc.org/indi/abide/)(42, 43). Functional images were subsequently preprocessed using the ABCD-HCP functional pipeline implemented via Docker (https://hub.docker.com/r/dcanumn/abcd-hcp-pipeline)(76). The preprocessing pipeline included head motion correction, spatial distortion correction, slice timing correction, coregistration to structural images, intensity normalization, and projection of fMRI signals onto the cortical surface. Surface-based rs-fMRI data were then resampled to the Human Connectome Project (HCP) derived fs_LR 32k surface mesh (https://github.com/Washington-University/Pipelines)(77).

Prior to functional connectivity gradient analysis, the preprocessed CIFTI dense time series were processed at the individual-participant level. For each participant, the time series at each cortical vertex and subcortical voxel were demeaned and variance-normalized, and all runs across sessions were then concatenated along the temporal dimension. The resulting preprocessed time series were then used for subsequent construction of functional connectivity matrices and gradient estimation.

Construction of cerebello-cortical functional connectivity and gradients

Cerebello–cortical functional connectivity (FC) matrices were constructed for each participant. First, the original pre-processed rs-fMRI data were spatially smoothed (Gaussian kernel with the FWHM = 6mm) using the Connectome Workbench command - *cifti-smoothing*. Cortical regions were defined using 1,483 regions of interest (ROIs) sampled on the fs_LR_32k surface with the fs_LR_900 mesh, following a previously established approach in which connectivity with these 1,483 ROIs was shown to provide a sufficiently rich representation of the functional connectivity profile of each voxel(37). Simultaneously, the time series from all cerebellar voxels were retained for subsequent connectivity estimation. We then computed Pearson's correlation coefficients between the time series of each cerebellar voxel (17,853 voxels) and each of the 1,483 cortical ROIs(45). The resulting individual correlation matrices were Fisher-transformed to z-values. Using an empirically motivated thresholding approach, only the strongest 10% of connections were retained to capture the most salient features of the connectivity profile (2, 24, 26, 78, 79).

To characterize the macroscale organization of cerebellar functional connectivity, we projected the high-dimensional connectivity matrices into a low-dimensional embedding space using the Laplacian Eigenmap algorithm(44). Pairwise affinity between cerebellar

voxels was estimated using cosine similarity between the corresponding rows of the FC matrix. Using the BrainSpace toolbox(44) (https://github.com/MICA-MNI/BrainSpace), we derived individualized functional gradients from the FC matrices. Subsequently, individual gradients were aligned across individuals using iterative Procrustes rotation(2). Given that the principal primary-to-transmodal gradient is closely associated with the neuronal microstructure and cognitive functions(24, 25), we primarily focused on ASD-related alterations in the principal gradient.

Since the individual-level analyses were performed across multiple acquisition sites, we harmonized the individual principal gradient to minimize potential site-related effects. Specifically, the ComBat harmonization method(80-82) was applied to the individual principal gradient values, with age and mean FD included as biological covariates. The harmonized principal gradient maps were subsequently used for individual-level statistical analyses.

Construction of within cerebellum functional gradients

To characterize intrinsic functional organization within the cerebellum, we constructed cerebellum-to-cerebellum FC matrices for each participant by computing Pearson's correlation coefficients between the time series of all pairs of cerebellar voxels (17,853 × 17,853). The correlation matrices were Fisher-transformed to z-values, and the top 10% strongest connections were retained. Pairwise affinity was then calculated using cosine similarity between the connectivity profiles of cerebellar voxels. Functional gradients were derived using the Laplacian Eigenmap algorithm implemented in the BrainSpace toolbox, following the same procedure as for the cerebello-cortical gradients. Individual gradients were aligned across participants using iterative Procrustes rotation. The principal gradient was used for subsequent analyses.

Meta-analytic cognitive associations with ASD-related gradient alterations

To characterize the cognitive functions associated with ASD-related alterations in cerebello-cortical principal gradients, we performed meta-analytic decoding using the Neurosynth platform (https://neurosynth.org/)(83). First, voxel-wise two-sample *t*-tests were conducted to assess group differences in principal gradient values between individuals with ASD and HCs, resulting in a cerebellar *t*-statistic map. Then, the statistical map was thresholded using false discovery rate (FDR) correction ($q < 0.05$) to retain only significant voxels and was subsequently separated into two contrast maps representing regions with significantly higher (ASD > HCs) and lower (ASD < HCs) principal gradient values in ASD(82).

For each contrast map, we used the *decoder* function implemented in Neurosynth to quantify its spatial correspondence with the precomputed meta-analytic activation map of each cognitive term in the Neurosynth database. Specifically, the *decoder* calculates the spatial correlation between the input statistical map and each term-specific meta-analytic map and converts the correlation into a standardized Z-score, which reflects the strength of the association between the observed spatial pattern and the corresponding cognitive function.

To facilitate interpretation and reduce nonspecific associations, the decoding analysis was restricted to a predefined set of cognitive and clinically relevant terms (742 terms),

following previously described approaches for manually filtering Neurosynth terms(84). Within this restricted vocabulary, only terms showing positive associations with nominal statistical significance ($p < 0.05$) were retained for subsequent interpretation.

Finally, the retained terms were visualized using word clouds, in which font size was proportional to the corresponding Z-score, such that larger words indicated stronger cognitive associations. Word clouds were generated separately for the ASD > HCs and ASD < HCs contrast maps to illustrate the cognitive functions associated with regions exhibiting increased and decreased principal gradient values in ASD, respectively.

Assessment of individualized cerebello-cortical gradient features

To characterize individual differences in cerebello-cortical functional organization, we computed two metrics for each participant: global gradient similarity and gradient dispersion.

Gradient similarity was defined as the spatial correlation between an individual's principal gradient map and the group-averaged principal gradient map. For each participant, we calculated the Pearson's correlation coefficient between their voxel-wise gradient values and the group-averaged map across cerebellar voxels. Higher correlation coefficients indicate greater similarity between an individual's gradient organization and the group-level topographic pattern.

Gradient dispersion was computed as the mean absolute deviation (MAD) of individual gradient values to capture the spatial variability of gradient organization. Specifically, for each participant, we calculated the average absolute deviation for each voxel's gradient value from the mean gradient across all cerebellar voxels. Higher dispersion values indicate greater heterogeneity or segregation of gradient values across the cerebellum, whereas lower values suggest more homogeneous or integrated organization.

To examine network-specific effects, we further quantified dispersion within each of six functional networks. Cerebellar voxels were parcellated according to a predefined 7-network parcellation scheme (excluding one empty network of limbic network)(45), and within each network, the MAD of individual gradient values was calculated separately for each participant.

Group differences in both similarity and dispersion metrics between ASD and HCs were assessed using two-sample *t*-tests, with significance threshold set at $p < 0.05$.

Support vector machine classification and statistical inference

To evaluate whether cerebello-cortical principal gradient features could discriminate individuals with ASD from HCs, we implemented a linear support vector machine (SVM) classifier using LIBSVM(85). Individual principal gradient maps (17,853 cerebellar vertices per participant) served as input features. To prevent potential data leakage, all confound adjustment procedures were performed within the training–testing framework. Specifically, for each training–test split, ComBat harmonization parameters were estimated exclusively from the training set and subsequently applied to the corresponding held-out test set to minimize site-related effects. Similarly, regression coefficients for age and mean FD were estimated using a general linear model implemented in SurfStat based

solely on the training data and then applied to both the training and test sets for covariate residualization. Sex was additionally included as a covariate for the ABIDE II cohort.

For each classification iteration, participants were randomly divided into a training set comprising 80% of ASD participants and an equal number of HCs, and an independent held-out test set consisting of the remaining participants. Within each training set, the optimal regularization parameter (*C*) was determined using five-fold cross-validation with a logarithmically spaced grid search ranging from 2^{-10} to 2^{10} (step size = $2^{0.2}$). A final linear SVM model was then trained using the optimal *C* and evaluated on the held-out test set. This entire procedure, including random train–test partitioning, hyperparameter optimization, model training, and testing, was repeated 100 times, and classification performance was summarized as the mean accuracy and area under the receiver operating characteristic curve (AUC).

To facilitate neuroanatomical interpretation of the classifier, the learned SVM weight vector was transformed using the Haufe approach(45, 86)

$$w_{\text{Haufe}} = w \cdot \text{Cov}(X_{\text{train}}),$$

which accounts for the covariance structure of the training data followed by L2 normalization. Positive weights indicate cerebellar vertices whose principal gradient values are positively associated with the ASD group, whereas negative weights indicate vertices more strongly associated with HCs. Haufe-transformed weight maps were averaged across the 100 classification iterations to obtain a stable estimate of the spatial contribution of each cerebellar vertex to the classifier.

Classification performance was evaluated using nonparametric permutation testing (1,000 permutations)(87). For each permutation, diagnostic labels were randomly shuffled while preserving class balance, and the entire classification pipeline, including model training, hyperparameter optimization, and evaluation, was repeated. The empirical *p* value was calculated as the proportion of permutation accuracies greater than or equal to the observed mean classification accuracy.

To further assess the cross-site generalizability of the classification model, we performed leave-one-site-out (LOSO) validation. In each LOSO fold, participants from one imaging site were completely held out as the test set, while participants from all remaining sites were used for model training. The held-out site was sequentially rotated across all available sites such that each site served as the test set once. Within each LOSO iteration, the optimal regularization parameter (*C*) was determined using five-fold cross-validation on the training data, followed by training of the final linear SVM using the selected *C*. Classification performance was then evaluated on the held-out site using accuracy and AUC.

Spatial correspondence analysis

The statistical significance of the spatial correspondence between the group-level principal gradient difference *t*-map and the mean Haufe-transformed weight map was assessed using Moran spectral randomization (MSR)(44). MSR generates surrogate spatial maps that preserve the spatial autocorrelation structure of the original data, thereby providing an appropriate null model for spatially correlated neuroimaging maps(88). A spatial weight

matrix was constructed from the Euclidean distances between cerebellar vertices in MNI152 space(45). Statistical significance (p_{moran}) was estimated as the proportion of surrogate correlations equal to or greater than the observed spatial correlation under permutation testing (1,000 permutations).

Behavioral phenotype associations with network-specific principal gradient

Because behavioral assessments differed across ABIDE sites, substantial missing data were present for several behavioral measures, particularly the Social Responsiveness Scale (SRS), which was unavailable for the ABIDE I cohort. Therefore, behavioral analyses were restricted to the ABIDE II cohort to maximize sample size while maintaining a homogeneous behavioral assessment framework (**Supplementary Table S3**).

Available behavioral measures were grouped into three clinically meaningful behavioral domains: (i) core autistic traits, including ADI-R social, verbal, nonverbal, and restricted/repetitive behavior (RRB) subscales, together with ADOS total, communication, social, and stereotyped behavior scores; (ii) restricted and repetitive behaviors (RRBs), including the ADI-R RRB subscale, ADOS stereotyped behavior score, and the SRS Autistic Mannerisms subscale; and (iii) social impairment, represented by all SRS subscales (total score, social cognition, social communication, social motivation, and autistic mannerisms). The complete classification of behavioral measures is provided in **Supplementary Table S4**.

As an exploratory analysis, three independent canonical correlation analyses (CCA)(89) were performed to examine multivariate associations between cerebello-cortical functional organization and behavioral phenotypes. For each CCA model, the brain variable set consisted of the mean principal gradient values within the six canonical cerebellar functional networks, including the visual (VN), somatomotor (SMN), dorsal attention (DAN), ventral attention (VAN), frontoparietal (FPN), and default mode (DMN) networks, whereas the behavioral variable set comprised the behavioral measures belonging to one of the three predefined behavioral domains. Only the first canonical variate pair was retained for subsequent analyses (*n_components* = 1). Canonical weights were examined to characterize the relative contribution of individual cerebellar networks and behavioral measures to each canonical mode.

Before CCA, all variables were standardized to *Z* scores using the StandardScaler. Participants with missing values in any brain or behavioral variable were excluded on a model-by-model basis(5), resulting in sample sizes of 39, 33, and 41 participants for the core autistic traits, RRB, and social impairment models, respectively.

Statistical significance was assessed using 10,000 permutation tests. For each permutation, participant labels in the behavioral matrix were randomly shuffled before refitting the CCA model, and the empirical *p* value was calculated as the proportion of permuted canonical correlations equal to or greater than the observed correlation. Because three CCA models were evaluated, *p* values were corrected using the Benjamini–Hochberg FDR, with significance defined as $q < 0.05$.

Transcriptomic analysis

Gene expression data were obtained from the Allen Human Brain Atlas (AHBA) dataset, consisting of 3,702 spatially distinct samples derived from six postmortem adult brains (age: 24–57 years; one female). Preprocessing and region-to-sample mapping were performed using the abagen toolbox(90) with default parameters unless otherwise specified. Probe-to-gene annotations were updated to the latest reference, and probes were filtered based on differential stability across donors (threshold > 0.5) to ensure robust regional expression patterns, followed by averaging of multiple probes per gene. The filtered samples were spatially mapped to a cerebellum-specific parcellation scheme, namely the NettekovenAsym32 atlas (32 asymmetric ROIs)(91) in MNI standard space. Given the inherent asymmetry of the cerebellar cortex, the bidirectional mirror-matching approach was disabled (lr_mirror = False); instead, a strict voxel-wise mask correspondence (exact = True) combined with a 5 mm spatial tolerance (tolerance = 5) was applied for sample-to-ROI assignment. To minimize inter-donor batch effects, expression values for each donor were first normalized across genes within each sample, followed by normalization across samples for each gene. The normalized samples were then aggregated across donors to derive a single mean expression value per gene for each ROI, yielding a final region-by-gene expression matrix (32 ROIs × 15,612 genes) for subsequent analyses.

To investigate the transcriptomic correlates underlying the observed neuroimaging-derived *t*-statistic maps, we performed partial least squares (PLS) regression between the *Z*-normalized gene expression matrix (predictor matrix **X**) and the *Z*-normalized regional *t*-statistics (response vector **Y**) using the *plsregress* function in MATLAB(92, 93). PLS1 was used to identify the principal low-dimensional pattern of gene expression that covaried with ASD-related alterations in cerebellar principal gradients. The statistical significance of the variance explained by PLS1 was assessed by randomly permuting the response variable 1,000 times. Bootstrap resampling was subsequently performed at the ROI level (1,000 iterations) to quantify the estimation error of each gene's PLS1 weight. For each bootstrap iteration, the PLS model was refitted and the signs of the resulting gene weights were aligned with the original weights by ensuring a positive correlation between the two weight vectors.

The bootstrap standard error (SE) for each gene's weight was estimated as the standard deviation across iterations, and a *Z*-score was calculated by dividing the original weight by its bootstrap SE. Following previous literature(94), genes with $|Z| > 2.58$ were identified as $PLS1^+$ (positive association) and $PLS1^-$ (negative association) gene sets.

The resulting $PLS1^+$ and $PLS1^-$ gene sets were separately submitted to the Metascape platform for automated functional enrichment analysis(95). All protein-coding *Homo sapiens* genes were used as the background set. Enriched biological processes and pathways were considered significant at FDR $q < 0.05$.

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Data availability

The raw and pre-processed data are available from the Autism Brain Imaging Data Exchange (ABIDE) I and II (http://fcon_1000.projects.nitrc.org/indi/abide/) and Allen Human Brain Atlas (AHBA, http://human.brain-map.org). The chosen participants' IDs and their behavior scales information of ABIDE I and II datasets are listed at our GitHub page (https://github.com/BIT-YangLab/ASD-Cerebellar-Functional-Gradients).

Code availability

The code used for data analysis is available on our GitHub page (https://github.com/BIT-YangLab/ASD-Cerebellar-Functional-Gradients). We used the ABCD-HCP functional pipeline via Docker for data preprocessing (https://hub.docker.com/r/dcanumn/abcd-hcp-pipeline). For genetic correlation estimations, we utilized the PLS genetic analyses available at https://github.com/SarahMorgan/Morphometric_Similarity_SZ. The visualization tool Metascape for genetic enrichment analyses is accessible at https://metascape.org/gp/index.html.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (grants number 82302175 and 62336002); the National Science and Technology Innovation 2030 Major Program and Project of China (grant number 2021ZD0200500 and 2021ZD0200506); the Beijing Municipal Science and Technology Commission (grants number Z171100000117012 and Z181100001518003). We thank the National Center for Protein Sciences at Peking University in Beijing, China for assistance on data analysis. This work was partially supported by Beijing Institute of Technology Kunpeng&Ascend Center of Cultivation.

Author contributions statement

T.Y., Y.Z., and G.Y. contributed to the conception of the study; W.Z. and G.Y. performed the data analyses; W.Z. and G.Y. created all visualizations; W.Z. and G.Y. wrote the first draft of the manuscript; and G.Y., R.Y., S.L., T.G., X.W., T.L., C.Y., T.Y. and Y.Z. reviewed and edited the manuscript.

Competing interests statement

All authors declare they have no competing interests.

Figures

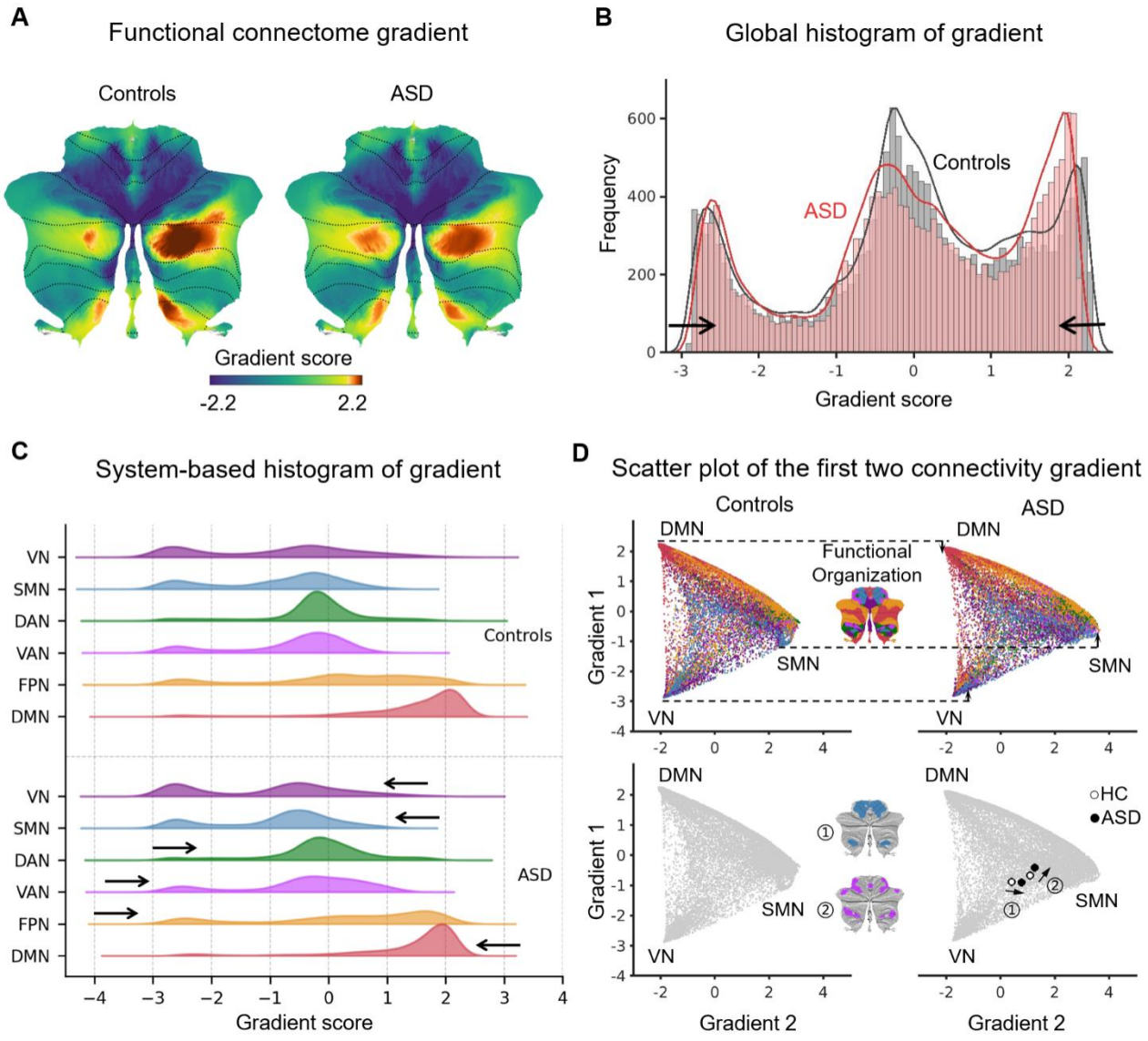


Fig. 1 | Cerebellar functional gradient mapping in healthy controls (HCs) and autism spectrum disorder (ASD). (A) Group-level principal gradient of cerebello-cortical functional connectivity in HCs and individuals with ASD. Color mapping follows a consistent scheme ranging from dark turquoise (unimodal regions) to sienna (transmodal regions). In both groups, the principal gradient exhibits a continuous organizational axis from sensorimotor to transmodal regions. Compared with HCs, individuals with ASD show systematically lower gradient values (lighter colors) within regions aligning with the default mode network. **(B)** Global histogram analysis of the principal gradient values in HCs and individuals with ASD. Relative to HCs, individuals with ASD show a contraction at the extreme high end of the gradient distribution. **(C)** Network-based histogram analysis revealed that individuals with ASD exhibited significantly lower gradient values in the DMN, SMN, and VN networks compared with HCs, whereas relatively higher gradient values were observed in the VAN, FPN, and DAN networks. Cerebellar voxels were assigned to six networks according to the cerebellar network parcellation defined by

Wu et al.(45) using an optimized winner-take-all strategy. **(D)** The upper panel shows scatter plot of the first two connectivity embedding gradients in HCs and ASD. Gradient 1 (y-axis) runs from primary sensorimotor to transmodal regions. Gradient 2 (x-axis) distinguishes somatomotor and auditory regions from visual regions. Scattered points are colored according to established functional communities for all six canonical networks. The bottom panel illustrates centroid shifts of functional networks in the two-dimensional gradient space. Specifically, the SMN centroid is displaced toward higher Gradient 2 values, whereas the VAN centroid shifts toward higher Gradient 1 and higher Gradient 2 values relative to HCs. VN, visual network; SMN, sensorimotor network; DAN, dorsal attention network; VAN, ventral attention network; FPN, frontoparietal network; DMN, default mode network.

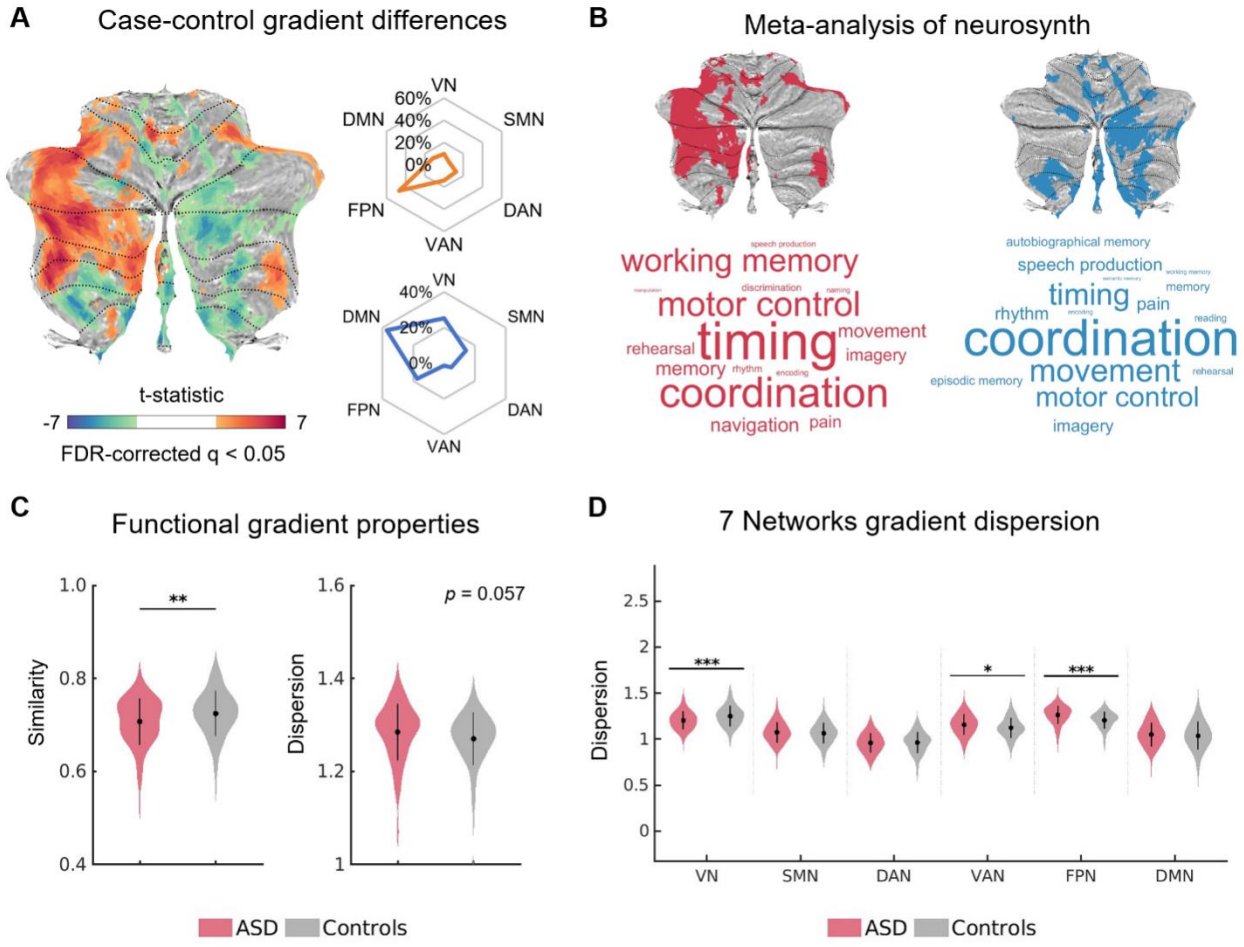


Fig. 2 | Statistical and organizational alterations of cerebellar functional gradients in HCs and ASD. (A) Voxel-wise statistical comparisons of cerebellar principal gradient values between HCs and individuals with ASD, as well as network-level distribution of mean percentage case–control differences across six cerebellar functional networks. Warm/cold colors represent significantly higher/lower gradient values in ASD relative to HCs. All statistical maps were thresholded at a voxel-level $p < 0.05$ with false discovery rate (FDR) correction $q < 0.05$ for multiple comparisons. **(B)** Word clouds of cognitive terms associated with cerebellar regions exhibiting significantly altered gradient values in ASD relative to HCs. Word clouds were derived from Neurosynth meta-analytic decoding based on the voxel-wise group-level *t*-map of the principal cerebellar gradient. The font size of each cognitive term corresponds to the meta-analytic *Z*-score generated by Neurosynth. **(C)** Differences in cerebellar functional gradient similarity to group-averaged template and dispersion between ASD and HCs. Two-sample *t*-tests were applied to compare these metrics between groups. **(D)** Network-specific gradient dispersion in ASD and HCs. Cerebellar voxels were partitioned into six canonical functional networks. Within each network, dispersion was calculated as the mean MAD of individual gradient values. Group comparisons between ASD and HCs were performed using two-sample *t*-tests with FDR correction $q < 0.05$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

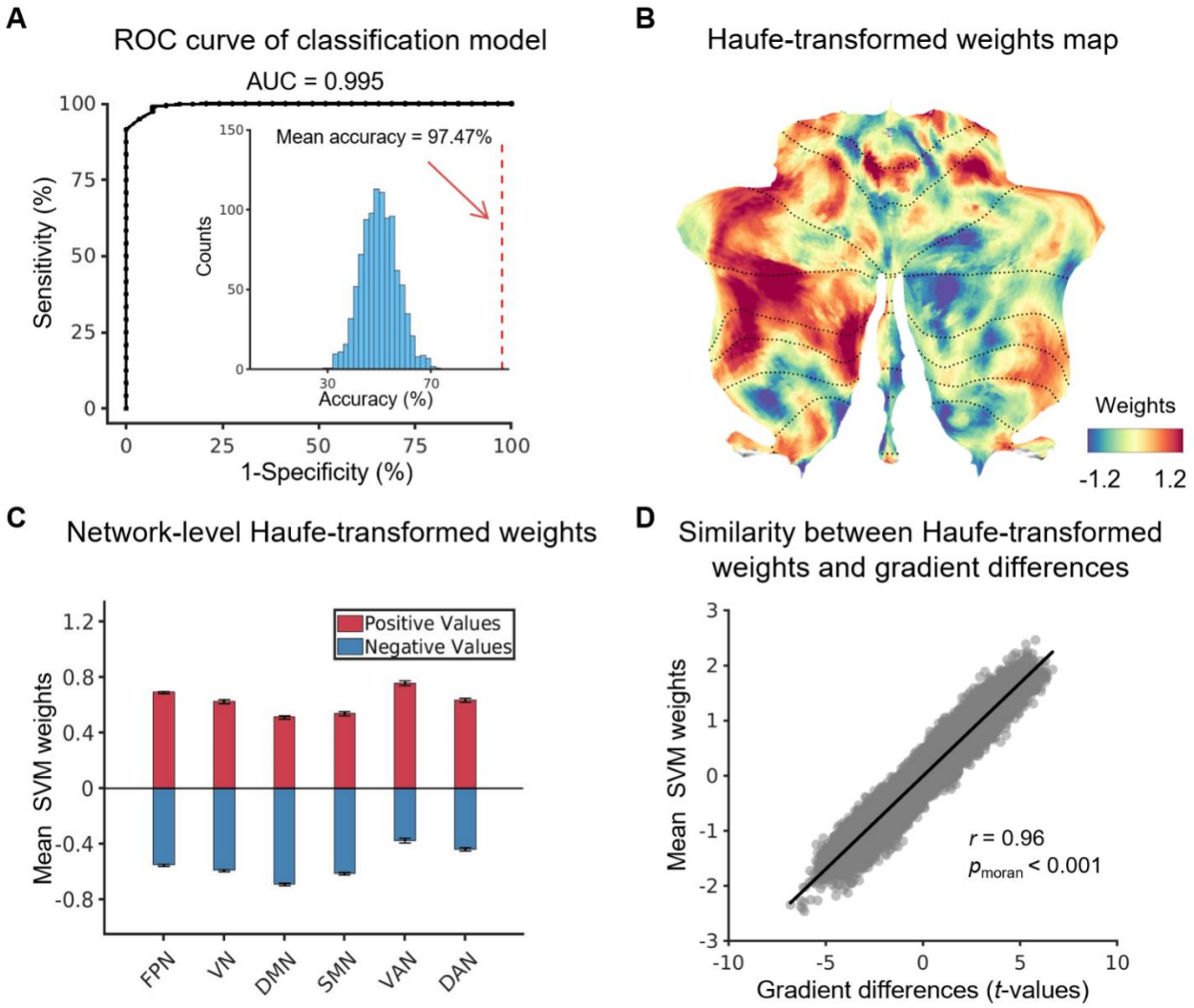


Fig. 3 | Classification of individuals with ASD and HCs using cerebellar principal gradient profiles via support vector machine (SVM). (A) Receiver operating characteristic (ROC) curve of the ASD versus HCs classification model. The area under the curve (AUC) is 0.995, with a mean classification accuracy of 97.47% (permutation test, $p < .001$). The inset histogram shows the null distribution of classification accuracies derived from 1,000 permutations, with the red dashed line indicating the observed mean accuracy. **(B)** Mean spatial map of Haufe transformed weights. Positive weights (red) indicate network regions whose gradient values are positively associated with the ASD group (i.e., higher values in ASD relative to HCs), whereas negative weights (blue) indicate regions positively associated with the HCs group (i.e., higher values in HCs relative to ASD). The weight map exhibits a heterogeneous spatial pattern across the cerebellum, with higher absolute values reflecting greater contribution to distinguishing ASD from HCs. **(C)** Bar plot of summed Haufe-transformed weights for each cerebellar functional network. Positive weights (red bars) and negative weights (blue bars) are shown, sorted in descending order by the sum of absolute values. **(D)** Spatial correspondence between the mean Haufe-transformed weight map and the group-level gradient difference *t*-map. A significant positive association was observed (Spearman's $r = 0.96$, $p_{moran} < 0.001$).

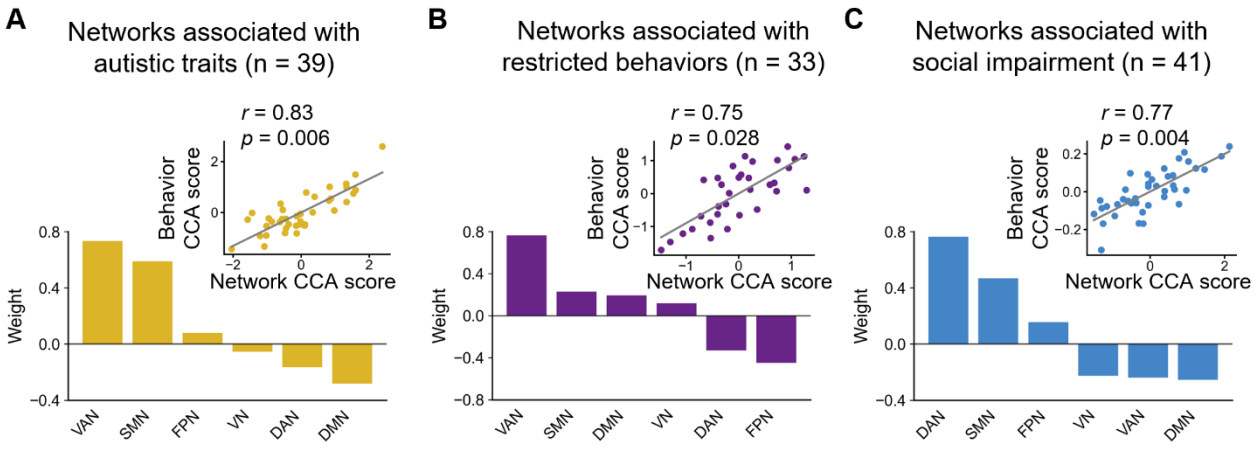


Fig. 4 | Three independent canonical correlation analysis (CCA) models revealed significant associations between principal gradient loadings and distinct autistic behavioral domains. **(A)** Networks associated with autistic traits (total sample, n = 39). The bar plot displays CCA weights for each cerebellar network, quantifying each network's contribution to the corresponding behavioral canonical mode. The scatter plot depicts the association between network CCA scores and behavioral CCA scores, with each dot representing an individual participant (r = 0.83, p = 0.006, q = 0.009). **(B)** Networks associated with restricted/repetitive behaviors (n = 33). The bar plot presents network-specific CCA weights, illustrating the contribution of each gradient-derived network to this behavioral domain. The scatter plot shows the correlation between network and behavioral CCA scores (r = 0.75, p = 0.028, q = 0.028). **(C)** Networks associated with social impairment (n = 41). The bar plot displays network-level contributions to the social-deficit-related CCA mode. The scatter plot depicts the relationship between network and behavioral CCA scores (r = 0.77, p = 0.004, q = 0.009). Positive correlations in scatter plots indicate that higher loadings on the latent canonical factor correspond to more severe autistic behavioral impairments. The magnitude of bar plot weights reflects the relative contribution of each cerebellar network to the canonical variate.

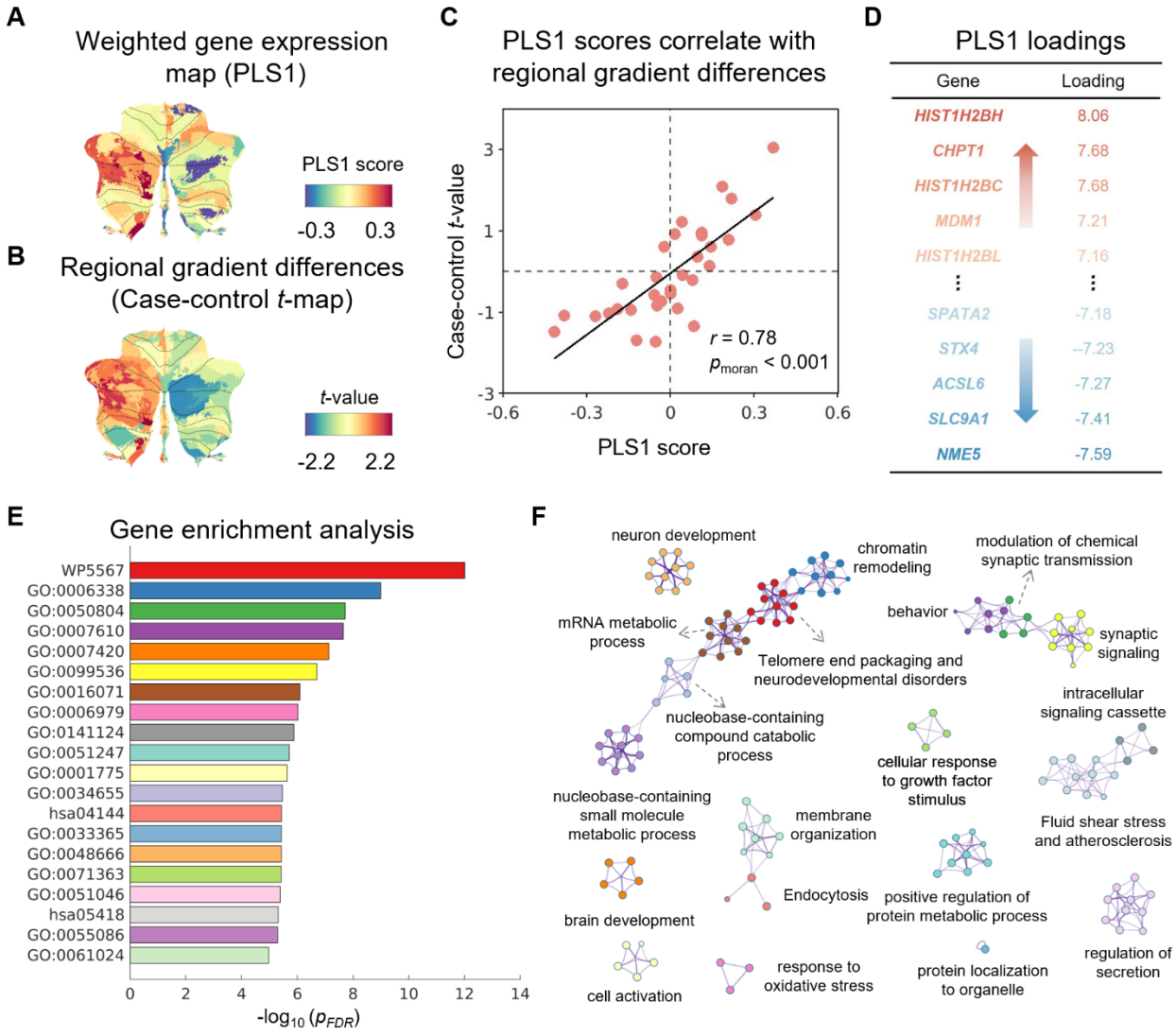


Fig. 5 | Transcriptomic correlates of ASD-related alterations in cerebellar principal gradient. (A) Spatial distribution of the first partial least squares component (PLS1) across the 32 cerebellar regions defined by the Nettekoven atlas(91). **(B)** Regional case–control *t*-statistic map of principal gradient differences projected onto the same cerebellar atlas. **(C)** Spatial correlation between regional PLS1 scores and regional gradient difference *t* values, with significance assessed using Moran spatial permutation ($r = 0.78$, $p_{moran} < 0.001$). **(D)** Genes ranked according to their PLS1 loading weights. The complete gene loading table is provided in Supplementary Data S1. **(E)** Functional enrichment analysis of PLS1^{+} associated genes. Bars represent significantly enriched Gene Ontology (GO) biological processes and pathway terms ranked by $-\log_{10}(p_{FDR})$. **(F)** Metascape enrichment network summarizing relationships among enriched biological processes. Nodes represent enriched terms, with node size proportional to gene number and colors indicating functional clusters.

Table 1 | Demographic characteristics of participants in the ABIDE-I and ABIDE-II cohorts.

ABIDE-I	ASD (*n* = 119, 119 males)	Controls (*n* = 124, 124 males)
Age in years (mean ± SD) [range]	18.77 ± 8.28 [7 – 50]	18.32 ± 7.32 [6 – 39]
Head motion in mm (mean ± SD) [range]	0.14 ± 0.06 [0.02 – 0.28]	0.13 ± 0.06 [0.05 – 0.30]
ABIDE-II	**ASD (*n* = 55, 48 males)**	**Controls (*n* = 54, 43 males)**
Age in years (mean ± SD) [range]	13.42 ± 6.22 [5 – 34]	14.61 ± 5.76 [5 – 26]
Head motion in mm (mean ± SD) [range]	0.13 ± 0.06 [0.05 – 0.27]	0.12 ± 0.05 [0.05 – 0.24]